

RAPID FLOOD MAPPING AND DAMAGE ASSESSMENT IN CAGAYAN RIVER BASIN USING SAR DATA AND GOOGLE EARTH ENGINE: A CASE OF TYPHOON ULYSSES

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ABSTRACT:

Philippines is one of the countries classified with severe vulnerability to climate change. Some of the Philippines' deadliest and most destructive typhoons happened just very recently. This study focuses on Typhoon Ulysses, a destructive category-4 typhoon that hit the Philippines on November 11, 2020. This study explores the use of Sentinel-1 SAR data and Google Earth Engine to perform rapid flood mapping and damage assessment in the Cagayan River basin. A change detection approach was employed to delineate flood extent which was then subsequently used to extract statistics on exposed population, affected croplands, and affected urban areas within the study area. The results showed that the total area of the flood extent in the Cagayan River basin on November 16, 2020 was 71,455 hectares. Meanwhile, 75,418 hectares of cropland were affected within the basin and the estimated urban area affected was 5,642 hectares. Approximately 61,730 people were also exposed to flooding. In summary, Cagayan and Isabela are the two most affected provinces within the Cagayan River basin. It was apparent that croplands/agricultural areas were affected more severely than urban areas.

1. INTRODUCTION

1.1 Background of the Study

1.1.1 Climate Change and Typhoon Ulysses

Climate change is the long-term shifts in the temperatures and weather patterns of the world (United Nations, n.d.). Such shifts happen naturally due to changes in the activity of the sun or large-scale volcanic activities. However, in recent times, anthropogenic activities have been the main driver for drastic changes in the climate (United Nations, n.d.). These activities typically release large amounts of greenhouse gases in the atmosphere which trap heat and raise temperatures. While a change in temperature of 1°C seems small, this number is already significant on a global scale. In fact, today, people around the world experience its effects in the form of increased frequency and magnitude of extreme weather-related events such as droughts, hurricanes, wildfires, and flooding (UNEP, n.d.).

According to National Geographic (2011), Philippines is one of the countries of the world classified with severe vulnerability to climate change. This means that although the country's carbon footprint is relatively small compared to the international giants, it still faces the brunt of climate change impacts. In an article published in 2023 on their website, the University of the Philippines – Diliman College of Science's Institute of Environmental Science and Meteorology (UPD-CS IESM) warns the public that stronger and potentially more destructive typhoons are inevitable because of climate change.

Some of the Philippines' deadliest and most destructive typhoons happened just very recently. Some examples include the Typhoon Haiyan in 2013, also known as Super Typhoon Yolanda which is considered as one of the most powerful tropical cyclones ever recorded. It claimed the lives of over 6000 people and

brought about \$2.2 billion worth of damages (Del Rosario, 2011; Del Rosario, 2014). There was also Typhoon Rai in 2021, also known as Super Typhoon Odette, which was the second most damaging typhoon ever recorded just behind Typhoon Haiyan, and Typhoon Rolly in 2020, internationally known as Goni, which was the strongest tropical cyclone to have made a landfall in the Philippines.

While there are several typhoon cases that may be studied, this study focuses on Typhoon Ulysses, also known as Vamco internationally. It was a very destructive category-4 typhoon that affected Vietnam and Philippines, causing the worst widespread flooding in Metro Manila since 2009. It hit the Philippines on November 11, 2020, with sustained winds of about 150 km/h and gustiness of up to 205 km/h (Patel, K., n.d.). Typhoon Ulysses affected a total of 1,268,170 families (equal to about 5,205,724 persons) in 7,287 barangays. The affected areas were in the National Capital Region (NCR), Cordillera Administrative Region (CAR), Regions I, II, III, IV-A, and IV-B (DSWD, 2021).

1.1.2 Remote Sensing and Synthetic Aperture Radar (SAR)

According to Lillesand (2015), remote sensing is a method of data collection about an object or area from afar. Specialized sensors are usually mounted on an aircraft or similar platforms such as a satellite. This advanced technology has revolutionized the way we monitor, study, and understand the surface of the Earth, its atmospheric composition, and oceans. One of the most common forms of remote sensing is Synthetic Aperture Radar (SAR). It uses radar signals to generate high-resolution images of the Earth's surface. SAR can penetrate clouds and is not heavily affected by weather conditions and time (Curlander & McDonough, 1991). Today, SAR has become an indispensable tool not only for environmental monitoring and natural resource management but also for disaster response and flood mapping.

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1.1.3 SAR and GEE in Flood Mapping and Damage Assessment

The ability of SAR to penetrate clouds and not depend on sunlight like other optical sensors make it particularly useful for monitoring areas prone to extreme weather disasters such as typhoons (Henderson & Lewis, 1998). SAR can be used to detect changes in land surface conditions before and after such events, providing necessary and valuable data and insights for damage assessment and recovery planning.

Integrating SAR with other modern advanced geospatial technologies like Google Earth Engine (GEE) enhances its capabilities for rapid, near-real time (NRT) flood mapping and damage assessment (Gorelick et al., 2017). GEE is a powerful computing platform that allows users to analyze large datasets for planetary-scale studies including SAR imagery. Combining SAR and GEE provides an avenue to identify flood extents and assess damages very quickly.

1.2 Significance of the Study

This study proves that geospatial technologies exist that can provide critical data that can be used for rapid damage assessment for large-scale flooding events. By enabling rapid assessment of affected areas, remote sensing workflows facilitate and allow timely, effective, and efficient response efforts which can reduce economic losses and potentially save lives. In addition to immediate disaster response, remote sensing can be used for long-term recovery efforts. Post-disaster assessments can identify most affected areas that may be prioritized for recovery. Lastly, historical remote sensing data contributes towards the understanding of the frequency and severity of typhoons and flooding, assisting in the development of more efficient and effective disaster mitigation strategies and response systems.

1.3 Research Objectives

This project has three objectives. First, this study aims to apply SAR data and GEE for rapid flood mapping in the Cagayan River Basin. Similarly, this aims to apply SAR data and GEE for damage assessment in the Cagayan River Basin. Finally, we aim to provide a qualitative evaluation of the resulting flood maps and damage assessments.

2. METHODOLOGY

2.1 Study Area

2.1.1 Cagayan River Basin

The Cagayan River Basin is the largest river basin in the Philippines. It is located in the northeastern part of Luzon. With a total land area of 27,493 square kilometers, it covers the provinces of Apayao, Benguet, Ifugao, Kalinga, Mountain Province, and Abra in CAR; Cagayan, Isabela, Nueva Vizcaya, and Quirino in Region II; and some parts of Aurora and Nueva Ecija in Region III (DENR, n.d.). Its river, the Cagayan River, is the longest and second largest in the Philippines, crucial in agricultural, fisheries, and hydroelectric power sectors. Bounded by Sierra Madre to the east, Central Cordillera to the west, Caraballo Mountain ranges to the south and Babuyan Channel to the north, its topography is varied from mountainous terrain to flat agricultural lands making it susceptible to various hydrometeorological hazards such as typhoons and flooding.



Figure 1 Extent of the Cagayan River Basin.

The Cagayan River Basin is often studied for flood dynamics and disaster impacts because of its sheer size and socio-economic importance. Using SAR and GEE to improve disaster preparedness and response is critical for this vulnerable region.

2.2 Data

The primary data used in this study was Sentinel-1 Ground Range Detected (GRD) data. Sentinel-1 GRD data is managed and distributed by the European Space Agency (ESA). Access to this data was done through GEE data catalog. Sentinel-1 data has a revisit time of 12 days and offers data in the C-band with dual polarization channels in VV and VH (Tumaneng et al., 2024). Two SAR images were used for study: one for the pre-flood analysis and the other for the post-flood analysis. The pre-flood SAR image used was dated October 23, 2020, while the post-flood SAR image used was dated November 16, 2020, which was the earliest available image immediately after Typhoon Ulysses hit the Philippines. The SAR images were in 10-meter resolution. The population data used in this study was a satellite-derived population layer named Global Human Settlement Layer (GHSL) which depicts the spatial distribution of population expressed as the number of inhabitants per cell in 100-meter resolution. For both cropland and urban areas, the 2020 European Space Agency (ESA) WorldCover data was used. This data has a 10-meter resolution and is based on Sentinel-1 and Sentinel-2 data. Auxiliary datasets for this study included the 30-meter Shuttle Radar Topography Mission (SRTM) Digital Elevation Model (DEM), the 30-meter JRC Global Surface Water Mapping Layers containing the location and temporal distribution of surface water up to year 2021 globally, and the Cagayan River Basin extent. Note that all of these datasets are directly available from the GEE Data Catalog except for the Cagayan River Basin extent which was downloaded from Geoportal PH.

Table 1 Summary of the datasets in this study

Dataset	Description	Source
SAR Images	Pre-flood and post-flood mosaics	Sentinel 1 (GRD) from GEE; 10m
Population Density	Exposed population	JRC GHSL: Global population 2020;100m
Cropland	Affected cropland	European Space Agency (ESA) WorldCover 2020; 10m

Urban Areas	Affected urban areas	European Space Agency (ESA) WorldCover 2020; 10m
Slope	Areas with >5% slope are excluded	Shuttle Radar Topography Mission (SRTM); 30m
Bodies of water	Exclude water bodies (>10 months)	JRC Global Surface Water Mapping Layers; 30m
Cagayan River Basin	Study area extent	Major River Basin Boundaries (Geoportal PH)

2.3 Flood Mapping

The flood mapping workflow used in this study follows the steps prescribed by the United Nations Platform for Space-based Information for Disaster Management and Emergency Response (UN SPIDER). The complete workflow is shown in Figure 2.

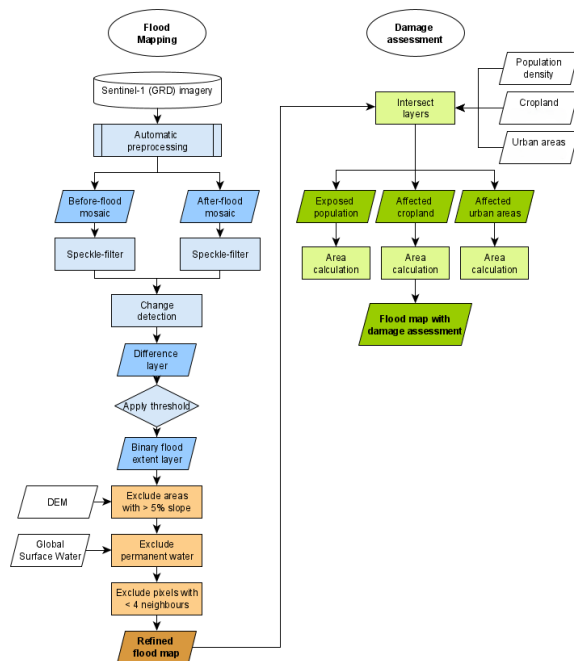


Figure 2 Rapid flood mapping and damage assessment workflow (UN SPIDER, 2019)

2.3.1 Change Detection

The pre-flood and post-flood SAR images used in this study were selected based on a set of parameters. These parameters were the spatial resolution, instrument mode, polarization, pass direction, and timeframe. For the spatial resolution, 10-meter resolution was chosen to provide the highest detail possible for the analysis. For the instrument mode, the Interferometric Wide swath (IW) mode was used because this is the instrument mode that fits this kind of study the most. For the polarization, the VH polarization was chosen because the UN SPIDER (2019) recommends this over VV because VV polarization might result in the overestimation of flood extent due to its higher backscatter from water surfaces. In addition, the VH polarization is more responsive to land cover changes, hence more suitable for this workflow (Tumaneng et al., 2024). The descending pass

direction was used in this study as it is the only available pass direction in the study area (see Figure 3).

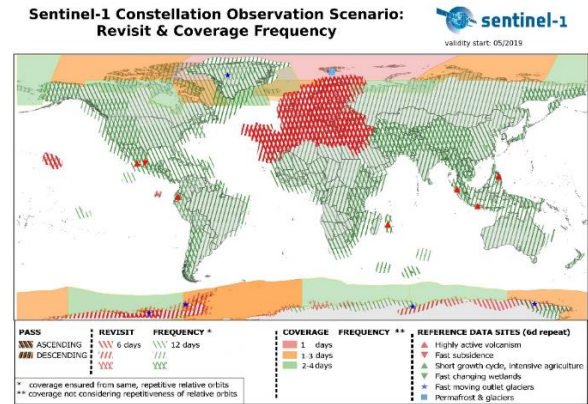


Figure 3 Revisit and coverage frequency of the Sentinel-1 constellation (ESA)

The pre-flood timeframe was set to be October 15-30, 2020, which returned the October 23, 2020 image. The post-flood timeframe, on the other hand, was set to be November 10-20, 2020, which then returned the November 16, 2020 image. After the two images were chosen, the speckle filter function with a smoothing radius of 50 was applied to both images to reduce the inherent noise in the images. It should be noted that the Sentinel-1 SAR GRD datasets hosted in and served by GEE have already undergone the following preprocessing steps (UN SPIDER, 2019):

- Apply-orbit-file (updates orbit metadata)
- ARD border noise removal (removes low intensity noise and invalid data on the scene edges)
- Thermal noise removal (removes additive noise in sub-swaths)
- Radiometric calibration (computes backscatter intensity using sensor calibration parameters)
- Terrain-correction (orthorectification)
- Conversion of the backscatter coefficient (σ^0) into decibels (dB)

To produce the flood extent layer, a simple, straight-forward change detection approach was employed. The post-flood mosaic was divided by the pre-flood mosaic. This resulted in a raster layer showing the degree of change per pixel. A threshold value of 1.25 was used to create a binary floor extent layer where the pixels with a value greater than 1.25 was assigned a new value of 1 and those with less than 1.25 was assigned a new value of 0.

2.3.2 Refining the Flood Map

To provide additional quality checks to the resulting flood layer extent and improve its reliability, additional datasets were used. First, the SRTM DEM data was imported to GEE as a new asset. The slope for the study area was calculated from this elevation data. Areas with slope greater than 5% were excluded because areas with steep slope cannot be flooded.

In addition, it is possible that the change detection approach to flood mapping produced false positives. To address this, the JRC Global Surface Water Mapping Layers dataset was used to exclude areas that have surface water for more than 10 months annually. This means that those areas are probably permanent inland waters as opposed to flooding caused by Typhoon Ulysses.

An additional step was also taken to ensure that noise is removed from the resulting flood extent layer. A connectivity evaluation was done to ensure that only the flood pixels with at least 8 neighbors or more are considered flooded areas. The pixels initially categorized as flooded but were found out to have less than 8 neighbors were considered noise and were removed from the resulting layer.

2.4 Damage Assessment

2.4.1 Exposed Population

To obtain an estimate of the number of people affected by the flood caused by Typhoon Ulysses, the 100-meter JRC Global Human Settlement Layer (GHSL) was used. The flood extent layer produced was then reprojected to the same coordinate system as the JRC GHSL layer. The resulting reprojected flood layer extent was then intersected with the JRC GHSL layer resulting in a new raster layer that contains only the cells with populations potentially affected by the flooding. The value of the cells indicates that number of individuals in that area. Hence, the total number of individuals affected by the flooding was calculated by taking the sum of all the cell values. The result was then displayed along with the other statistics.

2.4.2 Affected Cropland

To quantify the effect of flooding on cropland, the 2020 European Space Agency (ESA) WorldCover data was used. This raster layer contains multiple land cover types globally at 10-meter resolution. A masking function was written to mask all other land cover types except cropland. Similarly, the flood extent layer was reprojected to the same coordinate system as the ESA WorldCover data. The two layers were then intersected resulting in a new raster layer that contains only the cropland areas affected by flood. The area for each pixel was calculated (each pixel was 10x10m) and the sum for the areas of all the pixels was taken to record the total area of cropland affected by flooding.

2.4.3 Affected Urban Areas

Likewise, to quantify the effect of flooding on urban areas, the 2020 European Space Agency (ESA) WorldCover data was also used. This raster layer contains multiple land cover types globally at 10-meter resolution. A masking function was written to mask all other land cover types except the urban areas. Similarly, the flood extent layer was reprojected to the same coordinate system as the ESA WorldCover data. The two layers were then intersected resulting in a new raster layer that contains only the urban areas affected by flood. The area for each pixel was calculated (each pixel was 10x10m) and the sum for the areas of all the pixels was taken to record the total area of urban areas affected by flooding.

3. RESULTS AND DISCUSSION

3.1 Results

The primary output of this study is a map shown in GEE map canvas with key statistics resulting from the geospatial analysis performed above (see Figure 4).

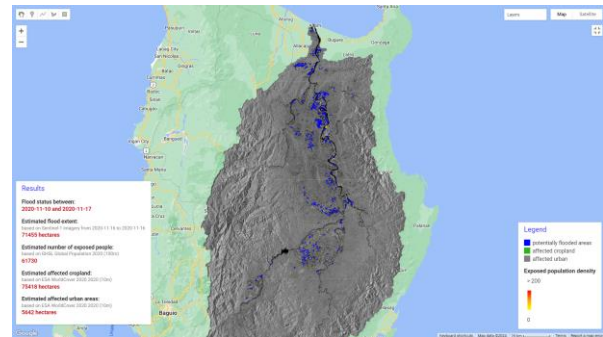


Figure 4 The analysis output in GEE map canvas.

To better visualize and for more control over the data, the resulting layers were exported to QGIS. Figure 5 shows the final flood map produced by the study.

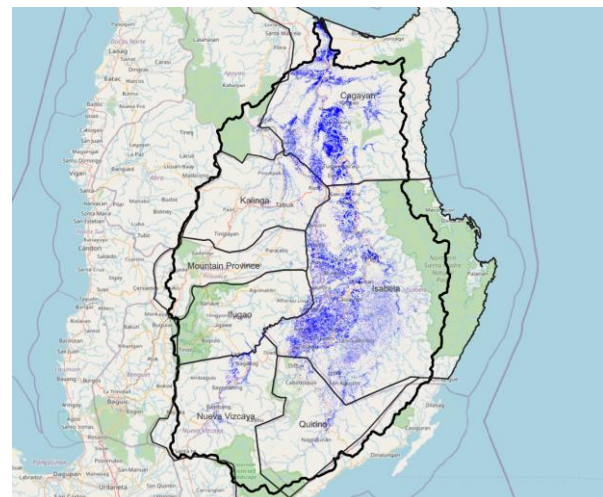


Figure 5 Resulting flood map for November 16, 2020.

The calculated total of the flooded areas delineated by this workflow was 71,455 hectares. In Figure 5, it can be seen that the majority of the delineated flooded areas are concentrated in Cagayan and Isabela (especially along the Cagayan River). Meanwhile, only some parts of Nueva Vizcaya, Quirino, and Kalinga were partially flooded.

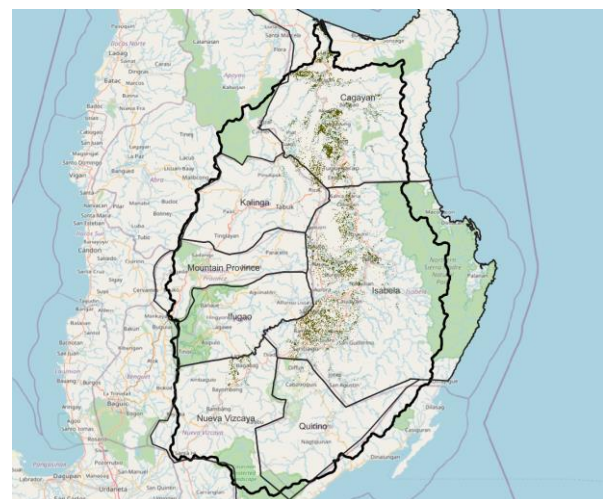


Figure 6 Flooded croplands for November 16, 2020

On the other hand, the estimated area of affected cropland is 75,418 hectares. What is notable is that the flooded cropland

areas are vast and the majority of which are found in Cagayan, Isabela, and some parts of Nueva Vizcaya. It can also be seen that most of these areas are near, if not literally adjacent to, the Cagayan River.

For the affected urban areas, the estimate is about 5,642 hectares mostly found along the sides of the Cagayan River. Especially when compared to the affected croplands, the affected urban areas are very sparse and fragmented making it difficult to see in a map relative to the scale of the analysis.

Finally, approximately 61,730 people were affected by flooding according to the data. Looking at the spatial distribution of the exposed population, those who live in municipalities or cities near the Cagayan River were affected the most.

3.2 Discussion

The estimated total flooded area calculated in this study was 71,455 hectares. However, Mata et al. (2022) in a similar study estimated the total flooded area at 62,087.9 hectares, a few thousands less than what this study has estimated. However, the study of Mata et al. did not exactly study the Cagayan River basin as a whole but only the provinces within it, excluding Nueva Vizcaya and Quirino.

Meanwhile, this study estimated that about 61,730 individuals were exposed to flooding. On the other hand, Mata et al. (2022) reported in their study that they estimated about 225,634 people were affected and DSWD DROMIC (2021) reports 925,210. This large discrepancy may be explained, however. The study by Mata et al. not consider the population's spatial variation as they were working on barangay boundary vectors, hence they had not access to the spatial distribution of the population within the barangay. Their method would most likely result in overestimation of the exposed population. On the other hand, DSWD's report on affected individuals was unclear. No definitions were made to explain how they arrived at such a number or what qualified one individual to be affected.

Unfortunately, we were unable to validate the reliability of the numbers for the affected cropland and urban areas as there are no similar studies that focus on the entirety of the Cagayan River basin.

This approach to rapid flood mapping and damage assessment using SAR data and GEE has its advantages and disadvantages. The following are the observed advantages:

- The study is replicable and can be adapted in any location. The workflow is easy and straightforward to implement anywhere in the world. Additionally, the methodology is flexible as only certain parameters (timeframe, instrument mode, polarization, and pass direction) are needed to be changed.
- The methodology also offers localization. Datasets used in this study can be swapped out for others if need be. This means that the methodology can be adapted for different geographic contexts if the data exists in the right format.
- The workflow is automated. This study makes use of GEE which requires certain level of programming. However, because the methodology is done purely using scripting, the workflow can be implemented without needed to redo the whole thing but only by tweaking some parameters of the script.

- Short waiting time. Because the workflow is already automated and the default datasets are readily available on the internet, the processing is relatively short and outputs can be obtained almost near-real time,
- The workflow is implemented on a cloud-computing platform. This means that it does not use the resources of your own computer. Machines that are not powerful enough can do the approach given a reliable internet connection.

Despite the advantages, there are also notable disadvantages in this approach. The disadvantages are as follows:

- False positives may be identified that are not caused by flooding.
- This approach has difficulties of detecting flood in urban or densely vegetated areas affecting its accuracy.
- The flood peaks (days where there are massive floods) may not be always captured due to the revisit time of the satellite. It is possible that the earliest satellite image available is already when the flood has subsided.
- The spatial resolution used in the workflow limits the accuracy of the statistics extracted from the study.

It is important to note the specific advantages and disadvantages of using SAR, GEE, and this rapid flood mapping and damage assessment approach. By understanding the pros and cons, we can continue to leverage its strengths and continuously work to address its limitations to further improve our results.

4. CONCLUSIONS AND RECOMMENDATIONS

4.1 Conclusions

This study has successfully demonstrated rapid flood mapping and damage assessment in the Cagayan River basin using Sentinel-1 SAR data and GEE. Satellite-derived flood maps were produced, and meaningful statistics were also extracted for the damage assessment.

The results showed that the total area of the flood extent in the Cagayan River basin on November 16, 2020 was 71,455 hectares mainly concentrated in Cagayan and Isabela. Some parts of Nueva Vizcaya, Quirino, and Kalinga were also partially flooded. Meanwhile, 75,418 hectares of cropland were affected within the basin, most of which are in Cagayan, Isabela, and Nueva Vizcaya as well. The estimated urban area affected was 5,642 hectares while approximately 61,730 people were exposed to flooding. In summary, Cagayan and Isabela are the two most affected provinces within the Cagayan River basin. It was apparent that croplands/agricultural areas were affected more severely than urban areas.

This study has proven that Synthetic Aperture Radar (SAR) and Google Earth Engine (GEE) can be used for rapid flood mapping and damage assessment. However, although accurate results are not guaranteed based on the high-level validation performed in this study, this method can still provide useful estimates. It is hoped that through this research, people learn that remote sensing techniques such as this provide alternative and useful ways of mapping and quantifying extreme disaster events that can help save lives.

4.2 Recommendations

While this study has successfully shown that rapid flood mapping and damage assessment can be done using Sentinel-1 SAR data and the cloud geospatial computing platform, GEE, there is still a lot of room for potential improvements.

First, the flood delineation approach used in this study is simple and straightforward. In some instances, this may not be as robust as needed. It is highly recommended that other flood mapping or change detection approaches are explored to improve the results of this study. Alternatively, given the same approach, it is best if future researchers can identify more accurate thresholds for the flood mapping through change detection workflow that would minimize the false positive results.

Because of time constraint and limitations on other resources, this study was conducted on a large scale. It is recommended that for future research, the granularity of the analysis (including the extraction of statistics) would be increased possibly at a barangay or municipality level to make the study more comprehensive.

Finally, this study only conducted a qualitative evaluation of the results. The best way forward is to find more robust ways of validation the results through quantitative means.

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